

Mandatory Assignment

Advanced Time Series Analysis

Department of Management and Economics

Student ID: 7301087

Contents

1	Introduction	2
2	Model and Simulation Setup	2
2.1	Model Specification	2
2.2	Simulation Design	2
3	Conditional Likelihood Construction	4
3.1	Log-Likelihood Contributions	4
3.2	Log-Likelihood Evaluation	4
4	Maximum Likelihood Estimation	4
4.1	Estimation Procedure	5
4.2	Small-Sample Results	5
4.3	Inference on the Autoregressive Parameter	6
4.4	Large-Sample Results	6
5	Quasi-Maximum Likelihood Estimation under Model Misspecification	6
5.1	QML under Model Misspecification	6
5.2	Small-Sample QML Results	7
5.3	Large-Sample QML Results	8
6	Comparing ML and QML Estimators	8
6.1	Simulation-Based Comparison	8
6.2	Interpretation	9
7	Conclusion	9

1 Introduction

This paper analyses the behaviour of maximum likelihood (ML) and quasi-maximum likelihood (QML) estimators in a time-series setting with time-varying volatility.

The analysis is conducted in a simulation framework, which allows the true data-generating process to be known and facilitates a comparison between correct specification and deliberate misspecification.

An autoregressive process with stochastic volatility is simulated and its parameters are first estimated by ML for both moderate and large sample sizes, allowing an assessment of finite-sample and asymptotic properties.

The same data are then analysed using a misspecified homoskedastic AR(1) model estimated via QML.

The paper concludes with a comparison of ML and QML estimators across repeated simulations.

The remainder of the paper is organised as follows. Section 2 introduces the model and simulation design, Section 3 presents ML estimation results, Section 4 discusses QML under misspecification, Section 5 compares ML and QML results, and Section 6 concludes.

2 Model and Simulation Setup

When working with standard ARMA models, constant variances are often assumed.

However, empirical macroeconomic and financial time series frequently display time-varying volatility, which motivates the introduction of a stochastic volatility structure.

This section presents the model specification and the simulation design used throughout the analysis.

2.1 Model Specification

The data-generating process is given by

$$y_t = c + \phi y_{t-1} + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t, \quad Z_t \sim \mathcal{N}(0, 1),$$

where the conditional volatility evolves according to

$$\sigma_t = \sqrt{\kappa + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2}.$$

2.2 Simulation Design

The model is simulated using the parameter values

$$c = 4, \quad \phi = 0.9, \quad \kappa = 0.75, \quad \alpha = 0.3, \quad \beta = 0.6.$$

Since $|\phi| < 1$, the autoregressive component is covariance-stationary and the unconditional mean is given by

$$E[y_t] = \frac{c}{1 - \phi}.$$

The simulation is therefore initialised at $y_0 = E[y_t] = 40$.

To initialise the volatility recursion, the unconditional variance of the innovation process is used,

$$\sigma^2 = \frac{\kappa}{1 - \alpha - \beta}.$$

A total of $T = 800$ observations is generated, with the first 50 discarded as burn-in, resulting in a final sample of $T = 750$.

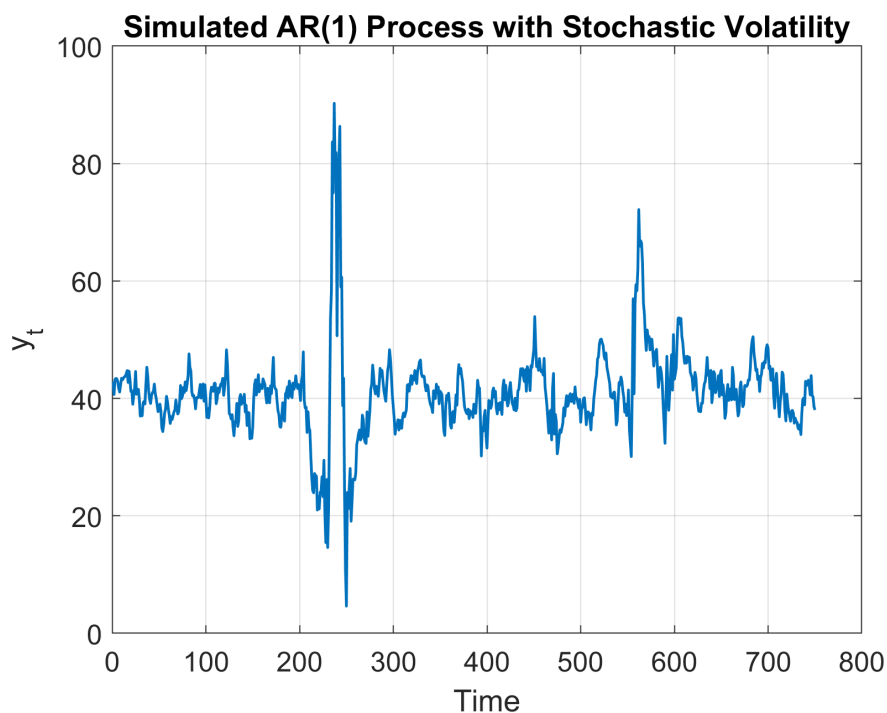


Figure 1: Simulated time series generated under stochastic volatility.

3 Conditional Likelihood Construction

This section outlines the construction of the conditional log-likelihood function for the stochastic volatility model introduced above.

The likelihood contributions are constructed by recursively recovering the innovation and the volatility process from the observed data.

3.1 Log-Likelihood Contributions

Given observations $\{y_t\}_{t=1}^T$, the innovation process is recovered as

$$\varepsilon_t = y_t - c - \phi y_{t-1},$$

with ε_1 set equal to zero.

Using the reconstructed innovations, the volatility sequence is obtained from

$$\sigma_t = \sqrt{\kappa + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2},$$

where σ_1 is initialised at the sample standard deviation of the innovation sequence.

Both ε_1 and σ_1 are excluded from the log-likelihood evaluation.

The conditional log-likelihood function is given by

$$\ln \mathcal{L} = \sum_{t=2}^T \left[\ln \left(\frac{1}{\sqrt{2\pi\sigma_t^2}} \right) - \frac{\varepsilon_t^2}{2\sigma_t^2} \right]. \quad (1)$$

Each term inside the sum represents the log-likelihood contribution at time t .

3.2 Log-Likelihood Evaluation

The conditional log-likelihood is evaluated for two alternative parameter vectors:

- (a) $c = 4, \phi = 0.95, \kappa = 0.75, \alpha = 0.3, \beta = 0.6,$
- (b) $c = 1.5, \phi = 0.75, \kappa = 0.5, \alpha = 0.6, \beta = 0.1.$

The resulting log-likelihood values are

$$\ln \mathcal{L}(\theta_a) = -1989.23, \quad \ln \mathcal{L}(\theta_b) = -2927.23.$$

Parameter set (a) yields a substantially higher log-likelihood value, which is expected since its parameters lie closer to the true data-generating process used in the simulation.

4 Maximum Likelihood Estimation

This section presents the estimation results obtained by maximum likelihood under correct model specification.

4.1 Estimation Procedure

The parameter vector

$$\theta = (c, \phi, \kappa, \alpha, \beta)'$$

is estimated by numerical maximisation of the conditional log-likelihood function described in (1).

Inference relies on the curvature of the log-likelihood and in theoretical terms, the covariance matrix of the ML estimator is given by

$$\text{Var}(\hat{\theta}_{ML}) = \left(-E \left[\frac{\partial^2 \mathcal{L}(y; \theta_0)}{\partial \theta \partial \theta'} \right] \right)^{-1}. \quad (2)$$

In practice, by the law of large numbers, the sample average of the observed second derivatives provides a consistent estimate of the theoretical expectation, so inference is based on the empirical Hessian evaluated at the optimum.

4.2 Small-Sample Results

For the sample of size $T = 750$, the maximum likelihood estimates are

$$\hat{\theta} = (5.436, 0.867, 0.631, 0.322, 0.599)'.$$

Standard errors are obtained from the inverse of the Hessian matrix described in (2), and 95% confidence intervals are constructed as

$$\hat{\theta}_i \pm 1.96 \cdot \widehat{SE}(\hat{\theta}_i).$$

The estimation results are reported in the following Table.

Parameter	Estimate	Std. Error	95% CI
c	5.436	0.712	[4.041, 6.830]
ϕ	0.867	0.017	[0.833, 0.901]
κ	0.631	0.151	[0.336, 0.926]
α	0.322	0.048	[0.228, 0.416]
β	0.599	0.046	[0.509, 0.689]

Table 1: Maximum likelihood estimates, standard errors, and 95% confidence intervals for $T = 750$.

The confidence intervals reported in Table 1 can be interpreted from two complementary perspectives.

From a data-generating process perspective, the intervals indicate that, under repeated sampling from the true DGP, the estimation procedure would include the true parameter values in approximately 95% of the samples.

In the present simulation, all true parameter values lie within their respective confidence intervals, suggesting accurate finite-sample performance of the ML estimator.

From a hypothesis-testing perspective, the 95% confidence intervals correspond to the set of parameter values that would not be rejected by a two-sided test at the 5% significance level, implying acceptance of a maximum Type I error probability of 5%.

4.3 Inference on the Autoregressive Parameter

Testing the null hypothesis $H_0 : \phi = 0.8$ yields a t -statistic of approximately 3.86.

This value exceeds the 5% critical threshold of 1.96, and so the null hypothesis is rejected at the 5% significance level.

The corresponding two-sided p -value is approximately 0.00011, indicating that, if $\phi = 0.8$ were true, observing a test statistic as extreme as the one obtained would be extremely unlikely. Since this probability is far below the 5% significance level, the null hypothesis is rejected.

4.4 Large-Sample Results

The estimation is repeated using a simulated sample of size $T = 49,950$.

In this case, the parameter estimates move extremely close to the true data-generating values, providing clear empirical evidence of consistency.

Parameter	Estimate	Std. Error	95% CI
c	4.027	0.075	[3.880, 4.174]
ϕ	0.899	0.002	[0.896, 0.903]
κ	0.760	0.025	[0.710, 0.810]
α	0.305	0.006	[0.293, 0.318]
β	0.594	0.007	[0.580, 0.609]

Table 2: ML estimates, standard errors, and 95% confidence intervals for $T = 49,950$.

A key change concerns inference: standard errors decrease substantially with $T = 49,950$, implying narrower confidence intervals and more precise estimates.

This occurs because the log-likelihood aggregates information over time (conditional contributions), so the Hessian of the log-likelihood function around the optimum increases with sample size; consequently, the inverse Hessian (and thus the covariance matrix) shrinks, leading to smaller standard errors.

5 Quasi-Maximum Likelihood Estimation under Model Misspecification

This section analyses parameter estimation under deliberate model misspecification.

The series generated under stochastic volatility is estimated using a homoskedastic AR(1) model with Gaussian innovations.

5.1 QML under Model Misspecification

The misspecified model is given by

$$y_t = c + \phi y_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma^2),$$

where σ^2 is treated as a constant parameter to be estimated.

The misspecification arises from both the omission of the stochastic volatility dynamics and the assumption of Gaussian innovations with constant variance.

As a result, the assumed conditional density does not coincide with the true data-generating process, and for this reason, quasi-maximum likelihood (QML) is introduced.

Unlike ML, QML does not require the log-likelihood function to be correctly specified: consistency can still be achieved even when the assumed model is misspecified. This makes QML particularly useful in settings where some aspects of the data-generating process, such as the volatility dynamics in the present case, are unknown or deliberately ignored.

QML operates by maximising a log-likelihood function that may not correspond to the true conditional density. Under such misspecification, the resulting estimators no longer converge to the true parameter vector, but instead to pseudo-true values (denoted by θ^*), defined as the minimisers of the discrepancy between the true model and the misspecified one.

A key property is that whenever a component of the model is correctly specified, its pseudo-true value coincides with the true parameter. In the present setting, although the volatility dynamics and distributional assumptions are misspecified, the conditional mean is correctly specified as

$$E[y_t | y_{t-1}] = c + \phi y_{t-1}.$$

Consequently, the QML estimators of c and ϕ converge to their true values, whereas the variance parameter σ^2 converges to a pseudo-true value.

Inference for the correctly specified parameters (here c and ϕ) remains valid under QML. However, because of misspecification, the Hessian-based ML covariance matrix is no longer appropriate.

Instead, the QML covariance matrix takes the robust form:

$$\text{Var}(\hat{\theta}_{QML}) = \left(-E \left[\frac{\partial^2 \mathcal{L}(y; \theta^*)}{\partial \theta \partial \theta'} \right] \right)^{-1} E \left[\left(\frac{\partial \mathcal{L}(y; \theta^*)}{\partial \theta} \right) \left(\frac{\partial \mathcal{L}(y; \theta^*)}{\partial \theta'} \right) \right] \left(-E \left[\frac{\partial^2 \mathcal{L}(y; \theta^*)}{\partial \theta \partial \theta'} \right] \right)^{-1}. \quad (3)$$

5.2 Small-Sample QML Results

Using the sample of size $T = 750$, the QML estimates of the conditional mean parameters are $\hat{c} = 5.254$ and $\hat{\phi} = 0.872$ and these estimates are close to their ML counterparts obtained under correct specification.

Parameter	Estimate	Std. Error	95% CI
c	5.254	1.984	[1.364, 9.143]
ϕ	0.872	0.050	[0.774, 0.969]
σ	3.687	0.292	[3.115, 4.259]

Table 3: QML estimates, robust standard errors, and 95% confidence intervals for $T = 750$.

Inference, however, differs substantially from the ML case (due to different covariance matrices).

The QML standard error of $\hat{\phi}$ is 0.0496, which is considerably larger than the corresponding ML standard error of 0.0173.

This increase reflects the effects of model misspecification, which reduce estimation efficiency. Furthermore, testing the null hypothesis $H_0 : \phi = 0.8$ yields a test statistic of approximately 1.45 and the null hypothesis is therefore not rejected.

Relative to the ML results, this illustrates how model misspecification increases the probability of failing to reject a false null hypothesis (increase of Type II error).

5.3 Large-Sample QML Results

The QML estimation is repeated using a much larger sample of size $T = 49,950$.

In this case, the QML estimates of c and ϕ move extremely close to both the ML estimates and the true data-generating values, providing clear empirical evidence of consistency.

Parameter	Estimate	Std. Error	95% CI
c	4.106	0.133	[3.846, 4.366]
ϕ	0.897	0.003	[0.891, 0.904]
σ	2.716	0.016	[2.685, 2.747]

Table 4: QML estimates, robust standard errors, and 95% confidence intervals for $T = 49,950$.

Even standard errors decrease substantially as the sample size increases, highlighting how, for the correctly specified parameters c and ϕ , the differences between the ML and QML estimators become asymptotically negligible.

6 Comparing ML and QML Estimators

This section compares the empirical behaviour of the ML and QML estimators across repeated simulations, for $T = 750$.

6.1 Simulation-Based Comparison

A total of $K = 500$ independent realisations of the stochastic volatility process is generated using the same parametrisation as in the baseline simulation.

For each realisation, the parameters of the correctly specified model are estimated by ML, while the homoskedastic AR(1) model is estimated by QML.

Kernel density estimates for the ML and QML sampling distributions of c and ϕ are then computed and displayed in Figure 2.

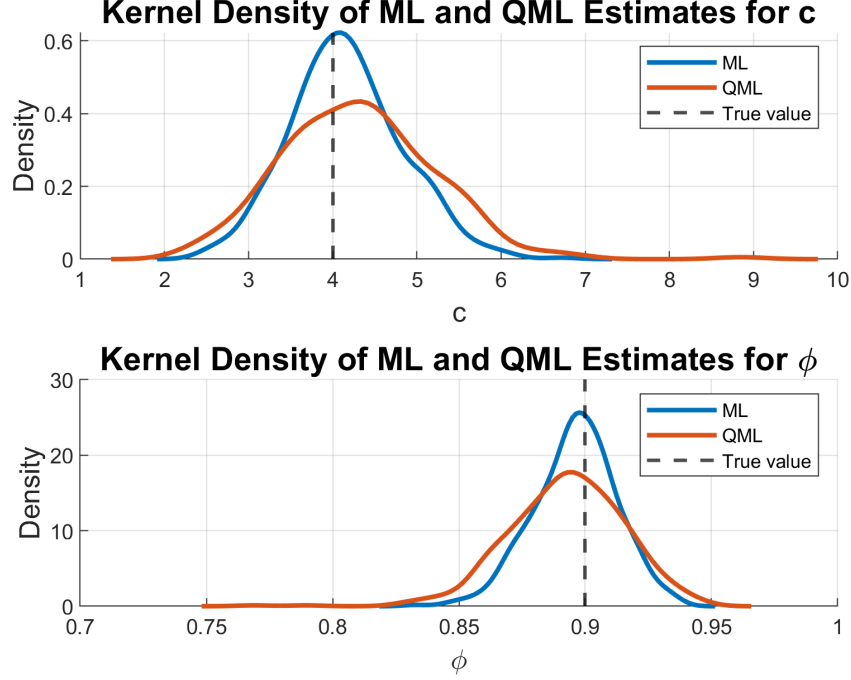


Figure 2: Kernel density estimates of ML and QML estimators for c and ϕ . The dashed vertical lines indicate the true parameter values.

6.2 Interpretation

In a finite-sample case, for both c and ϕ , the ML sampling distributions are tightly concentrated around the true parameter values, reflecting consistency and high efficiency under correct specification.

The QML sampling distributions are instead slightly shifted relative to the truth and visibly more dispersed, indicating the presence of a small bias and a loss of efficiency (caused by model misspecification).

7 Conclusion

The analysis showed that, in a setting where the conditional mean is correctly specified but the volatility dynamics are not, both ML and QML recover the mean parameters reliably. For a finite-sample case, ML delivers the most precise estimates, while QML exhibits a modest loss of efficiency and a small bias; however, as seen before, these distortions diminish quickly with sample size.

Beyond this specific application, the results reflect the broader role of QML as a robust estimation method under partial model misspecification. Even when certain components of the data-generating process are incorrectly specified or deliberately simplified, QML still delivers valid inference for the correctly specified parameters through its robust covariance matrix. This makes QML a practical and dependable tool in empirical time-series analysis, where full knowledge of the underlying stochastic structure is rarely available.